

Chia-Hung Kuo

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Technical Skills

Programming: Python, SQL, Stata, R

ML/DS Libraries: scikit-learn, PyTorch, XGBoost, pandas, Polars, EconML, CausalML

Visualization: matplotlib, seaborn, Tableau, Power BI

MLOps: MLflow, Optuna, Git, Docker, AWS, Terraform

Data Engineering: Spark (PySpark), Airflow, dbt, Kafka

Data Platforms: MySQL, BigQuery, Redshift, Databricks, Snowflake

Projects & Working Papers

A/B Testing with Uplift Modeling and Causal Decomposition

11/2025

- Implemented S/T/X/R meta-learners achieving 0.1753 Qini on 13.9M user RCT; decomposed CATE via instrumental variables to enable differentiated bidding strategies
- Quantified tradeoff: 10% ranking cost for strategic bid optimization

Multi-Target Stacking: A Novel Ensemble Framework

09/2025

- Developed a new ensemble framework using auxiliary targets with 57% improvement over standard stacking on 14M+ financial time series records
- Validated across 12 walk-forward folds maintaining stable 95.6% PR-AUC despite distribution shifts

Family Structure and Young Adult Child's Geographic Mobility

03/2023

- Built structural discrete choice model on NLSY97 panel data (~9,000 individuals) using maximum likelihood estimation to decompose migration decisions into preference and cost components
- Identified 12% reduction in migration probability for two-parent households, driven by 35% higher home premium rather than financial constraints

Compulsory Schooling and Locus of Control: Evidence from China

02/2021

- Applied difference-in-differences IV strategy exploiting China's 1986 Compulsory Schooling Law variation across 25 provinces (16,500+ individuals, CFPS data)
- Estimated 0.84 SD increase in internal locus of control from completing junior high school, with effects concentrated among policy compliers (females, rural residents)

Experience

Taiwan Assessment and Evaluation Association, Taipei, Taiwan

Data Science Consultant

09/2024-Current

- Built time series regression and latent variable model for anomaly detection across 1,600 department assessments annually, identifying outlier patterns in national higher education accreditation

Michigan State University, East Lansing, Michigan

Graduate Researcher & Teaching Assistant

09/2018-06/2024

- Conducted causal inference research (IV, DiD, MLE) on large-scale public surveys
- Excellence in Teaching Award (2022); Honorable Mention, Peter Schmidt Award (2019)

Institute of Labor Economics (IZA), Bonn, Germany

Research Assistant

01/2017-07/2018

- Supported GLM|LIC program with research coordination and publication workflows

Education

PhD Candidate (ABD) in Economics, Michigan State University, East Lansing, MI

06/2024

MSc in Economics, University of Bonn, Bonn, Germany

06/2018

BSc in Quantitative Finance, National Tsing Hua University, Hsinchu, Taiwan

06/2014